Resolution Geometry

Crossing into Physics: The Consilience Engine and the Physical-Curvature Programme

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Abstract

No framework computes physical values free of assumptions [1, 3]. Every empirical claim is tested together with its auxiliaries (Duhem–Quine), and no finite procedure certifies against an undeclared class of alternatives (Le Cam; ladder L56). This paper shows what can be done instead, and that it can be automated.

A *physical value* is certified as a value and an uncertainty *conditional on a computed residual set of assumptions*. The certificate comes from independent routes whose remaining assumptions are disjoint, and it uses four tools:

- invariance across protocols;
- an assumption-overlap matrix;
- robustness over a declared model class;
- commitment blinding with an invariant statistic.

When routes form contradicting consistent clusters, the engine returns a CORRESPONDENCE. It resolves the correspondence only by pre-registered tests of each cluster's distinctive assumptions, using observables that enter neither estimate.

On real GPS data from two GEONET stations, four routes to the same baseline length split into two clusters:

- *code:* undifferenced and double-differenced code;
- *phase and published:* double-differenced carrier phase and the published GEONET coordinates.

In round 1 both clusters survived, and the registered expectation that the code test would fail was wrong. That test had no power against a constant bias. Round 2, registered before it was run, falsified the code cluster's whiteness assumption with per-satellite code biases of 0.14–0.16 m (z up to 9). The phase cluster's integer-ambiguity assumption had already survived an independent test with chance probability about 10^{-4}.

The result is the baseline 3335.3913 ± 0.0034 m, LOCKED conditional on five listed assumptions. Three canaries are caught: unblinding before freezing, pretending the falsified test passed, and preferring a cluster without evidence. A resolution ladder types physical-curvature effects as detected or STRUCTURAL WALL[τ] and opens RG's physics programme.

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1 The boundary, stated exactly

No-Go 1.1 (No assumption-free physical value).

- Every route to a physical quantity is conditional on an assumption set (Duhem–Quine [4]).
- Certification against all alternatives is impossible without a declared class (Le Cam [5]; ladder L56).
- Some conditioning is irreducible. Conventions (SI definitions) and framework theories (relativity in the clock terms) are tested by the whole of physics, not by one experiment.

The task is therefore not to remove assumptions. It is to make the passage from fit to value a computed object, with an explicit residue.

Fitting versus physical determination. A fit chooses parameters in one model to match one dataset. A value becomes physical through three properties, none of which is a better fit:

1. it predicts other phenomena;
2. it is invariant across protocols;
3. routes with different assumptions agree, as in Perrin's determinations of Avogadro's number [6].

RG already formalizes the second (descent and Γ_{obs}; in the chaos frontier, the slope is physical and the intercept instrumental). This paper mechanizes the third.

2 The consilience engine

Definition 2.1 (Route)**.** A route R_i is a protocol together with its assumption set A_i. It returns $v_i \pm u_i$ for the target quantity.

Theorem 2.2 (Consilience certificate)**.** *Let S be a set of routes that agree pairwise ($|v_i - v_j| \le k\sqrt{u_i^2 + u_j^2}$). An undeclared error that shifts the routes of S by amounts whose pairwise differences exceed that bound is detected. In particular, an error in an assumption $a \notin A_j$ for some $j \in S$ leaves v_j unchanged, so it is detected once its effect on another route of S exceeds the bound. An error that escapes detection either shifts every route equally, or lies in $\bigcap_{i \in S} A_i$, or is below resolution.*

Proof. Agreement is a condition on pairwise differences, and a shift common to all routes leaves those differences unchanged. An assumption absent from A_j does not enter route j. □

Proposition 2.3 (Clusters and correspondences)**.**

- *The engine returns every maximal pairwise-consistent cluster.*
- *If two clusters contradict each other, the status is CORRESPONDENCE.*
- *The correspondence is resolved only by tests of each cluster's* distinctive *assumptions (shared inside the cluster, absent outside it). Such a test must be registered before it is run, must use observables that enter neither cluster's estimate, and must have power against the alternative in question.*

- *Sequential tests are allowed, provided the conditioning is reported.*

Proposition 2.4 (Commitment blinding)**.** *If the reported statistic is invariant under a group G, blinding the reference by a random $g \in G$ costs nothing and hides every non-invariant feature. A commitment* hash(g) *published before the results are frozen certifies the order commit $\rightarrow$ freeze $\rightarrow$ unblind externally. This is RG's version of blind analysis [8].*

Proposition 2.5 (Model-class robustness)**.** *Recompute each route over the declared alternatives. The route is FORCED if the spread is within its own uncertainty, and CONDITIONAL otherwise. This is the uncertainty-geometry certificate applied to modelling choices, and it is bounded by L56: robustness is always relative to the declared class.*

3 The demonstration on real GPS data

The target is the baseline length between GEONET stations 0759 and 3040, recorded on 2 April 2005 from 00:00 to 01:00 GPST. This is the same data as in the companion paper *Real Measurements* [2], with relativistic terms as in [9].

route	method	v (m)	u (m)
R1	undifferenced code positions (no anchor)	3335.7324	0.0796
R2	double-differenced L1 phase, float ambiguities	3335.3903	0.0035
R3	published GEONET coordinates (independent pipeline)	3335.4252	0.02 (declared)
R4	double-differenced code	3335.5615	0.0284

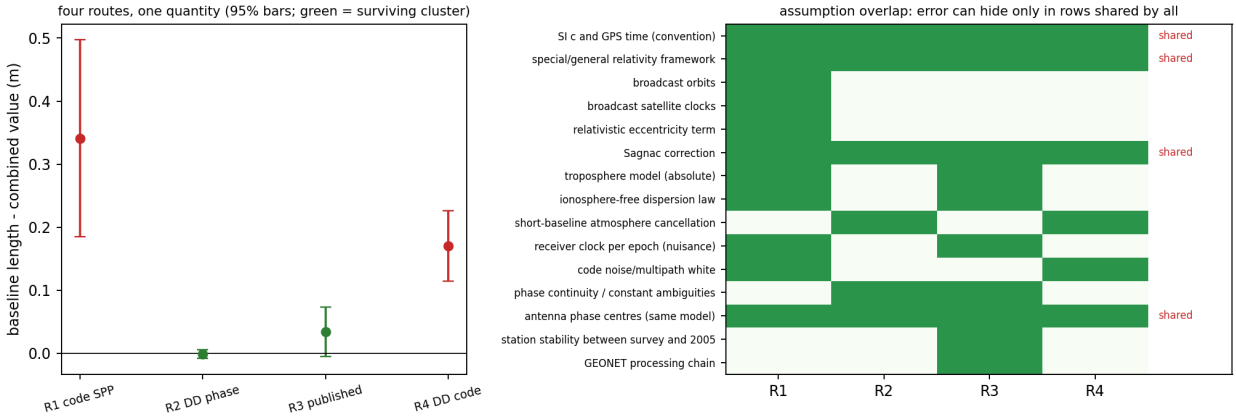


Figure 1: Left: four routes to one quantity; green marks the surviving cluster. Right: the declared assumption matrix. An undetected error can hide only in rows shared by every route of the consistent cluster.

Clusters. The pairwise z values are:

- R1–R4: 2.0 and R2–R3: 1.7 (consistent pairs);
- R1–R2: 4.3, R1–R3: 3.7, R2–R4: 6.0 and R3–R4: 3.9 (contradicting pairs).

There are two contradicting clusters, so the status is CORRESPONDENCE. Their distinctive assumptions are "code noise and multipath white" for the code cluster and "phase continuity, constant ambiguities" for the phase cluster.

Round 1 (registered before it was run).

- *Phase.* All five float double-difference ambiguities are within 0.08 cycles of an integer, with standard deviations up to 0.05 cycles. The chance probability is about 10^{-4}. Fixing them to integers moves the baseline by 4.6 mm [7]. Count rigidity (L8) is visible in real data.
- *Code.* The 10-minute block-mean ratio is 0.93, inside the whiteness band (1.60).

The registered expectation (code fails) was wrong. The test cannot detect a *constant* bias, the alternative in question. Both clusters survived, and the engine kept CORRESPONDENCE.

Round 2 (registered before it was run, conditional on round 1). The per-satellite mean of double-differenced code minus integer-fixed phase gives 0.140, 0.150, −0.058, 0.163 and 0.163 m, with z values of 4.1, 6.4, 1.4, 5.9 and 9.1. Constant code biases of decimetres falsify the code cluster's distinctive assumption. Their common sign points to the reference satellite's code multipath. The correspondence resolves by rule.

Result.

baseline = 3335.3913 ± 0.0034 m, LOCKED (conditional) on: SI c and GPS time (convention); relativity framework; Sagnac correction; phase continuity and integer ambiguities; antenna phase-centre model.
Rejected: {R1, R4}, because of a constant per-satellite code bias (round 2, conditional on round 1).
Declared u_{R3}: the critical margin is 0.011 m; below it, R2 and R3 would disagree.

Supporting checks.

- *Injected errors.* A 0.2 m error injected into R2's phase-only assumption is detected ($z = 11.6$). A common scale error in the intersection is provably undetectable.
- *Model-class robustness.* The phase route spreads by 5.5 mm across five declared alternatives (CONDITIONAL at its 3.5 mm). The code route spreads by 18 mm (FORCED at its 28 mm).
- *Blinding.* The statistic is rotation-invariant, the hidden direction is offset by 136°, and the commit, freeze and unblind order is verified. A pipeline that unblinds early is caught.

4 Registration record

In this demonstration the engine caught its author four times.

1. The first registration claimed that all routes agree. They do not.
2. A second version hard-coded the "right" cluster, {R2, R3}. The subset routine then picked the other cluster and the automatic diagnosis blamed the phase assumption. This exposed the hard-coding as smuggling, and it was replaced by the cluster and correspondence rule.
3. The round-1 expectation (code fails) was wrong, because the test lacked power against a constant bias.
4. An injection meant to show detection moved R2 *toward* R3, so it tested nothing.

Each correction changed the claim to match what the data support, or added a test with power against the alternative. None relaxed a threshold.

5 Physical curvature: the resolution ladder

effect	size (m)	resolved by	status
Sagnac (rotating slice, L105)	24.1	code	DETECTED
relativistic clock eccentricity (duration layer)	5.3	code	DETECTED
Shapiro delay, one station	0.014	absorbed by clock nuisance	declared model
Shapiro delay, double difference over 3.3 km	4×10^{-6}	none	STRUCTURAL WALL[3.5 mm]
chord minus arc of the ellipsoid over the baseline	4×10^{-5}	none	STRUCTURAL WALL[3.5 mm]

Proposition 5.1 (Curvature is typed by resolution)**.** *A curvature effect is DETECTED by a route iff its route-level magnitude exceeds k u. Otherwise it is a STRUCTURAL WALL[u] (L114). A longer baseline, a better route or a new protocol moves the effect across its wall, and RG states which.*

The rotating-frame curvature of the slice (Sagnac holonomy) and the gravitational clock effect are detected in this data. Spacetime curvature in signal propagation (Shapiro) is centimetric at one station, but cancels to micrometres over 3 km, far below the phase resolution. This is not a failure. It is the exact location of the next experiment.

6 The programme: RG physics, opened

1. *Physical-value certificates.* Each route is a protocol with a declared assumption set. The certificate reports the value conditional on the intersection over the surviving cluster, together with the rejected clusters and their falsified assumptions.

2. *An assumption ontology.* The assumption matrix is itself a declaration. A missing row shared by all routes would hide an error, by Theorem 2.2. Each entry needs a justification, and the matrix must be auditable and extensible.

3. *Power-aware tie-breaking.* A test that cannot detect the relevant alternative decides nothing. Power analysis belongs in the registration.

4. *Equivariant blinding as standard.* Choose the invariant statistic first, then blind by its group, with the commitment published externally.

5. *The physical-curvature ladder.* The next walls to cross are known:

   - Shapiro delay on long baselines, very-long-baseline radio interferometry (VLBI) and lunar laser ranging;
   - frame dragging, through satellite laser ranging;
   - tidal and loading deformation, through long time series.

   Each is a curvature effect moving across a STRUCTURAL WALL[τ] as routes improve.

6. *Observable versus physical curvature.* RG separates the curvature of the observable connection (L76), the curvature of the latent connection under declared covariance (L80), and physical curvature measured as a quantity invariant across protocols. Physical curvature is the third kind. Here it was identified by protocol invariance (Sagnac at both stations, with and without differencing) and certified by consilience.

What can and cannot be automated. The engine automates the bookkeeping:

- route certificates and the overlap matrix;
- clusters and correspondences;
- registered tie-break tests;
- blinding and status.

It cannot create new independent routes. These cost experiments, not computation. The proof standard it meets is "conditional on S, with S listed and each member's support cited". That is the metrologist's standard, and it is the strongest an empirical science can state.

7 Non-claims

- This is one baseline from one hour.
- The assumption matrix is a declaration.
- R3's uncertainty is declared, and its critical margin is reported.
- Round 2 is conditional on round 1.
- Within a single script, the blinding demonstrates the protocol. Its force requires the commitment to be published externally.
- No new physical constant is claimed.

A Reproducibility

`rg_consilience_attack.py` (CN01–CN11) passes 11/11 on the bundled RINEX data. It imports the real-data pipeline from *Real Measurements*. Its three canaries are caught:

- unblinding before freezing;
- pretending the falsified test passed (the status reverts to CORRESPONDENCE);
- preferring a cluster without evidence (the decision is unsupported by the recorded tests).

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